

on vectors and matrices. SageMath provides an excellent environment for linear algebra by integrating highly optimized libraries such as BLAS, LAPACK, ATLAS, LinBox, and IML, enabling efficient and scalable matrix computations.

```
sage: A = matrix([[1,2],[3,4]])
sage: A.det()
-2
sage: A.eigenvalues()
[-0.3722813232690144?, 5.372281323269015?]
sage: Ai=A.inverse()
sage: Ai
[ -2   1]
[ 3/2 -1/2]
sage: A*Ai
[1 0]
[0 1]
sage: Ai*A
[1 0]
[0 1]
```

Figure 4: A few SageMath one-liners for linear algebra

Let us now understand the output produced by the linear algebra functions shown in Figure 4. The function `det()` computes the determinant of a matrix. In this example, the determinant of the matrix A is -2 , indicating that the matrix is non-singular (its determinant is non-zero) and hence invertible. The function `eigenvalues()` computes the eigenvalues of the matrix, which are approximately -0.3723 and 5.3723 . These values satisfy the characteristic equation of the matrix and have numerous applications in fields such as machine learning, computer graphics, signal processing, and the solution of systems of differential equations. The function `inverse()` computes the inverse of the matrix A , denoted by A^{-1} . Finally, the matrix products $A \cdot A^{-1}$ and $A^{-1} \cdot A$ both yield the identity matrix, confirming that A^{-1} is indeed the inverse of A . This follows directly from the defining property of the inverse of a matrix: $AA^{-1} = A^{-1}A = I$, where I denotes the identity matrix.

Calculus

Calculus is the branch of mathematics that studies change and accumulation through the concepts of limits, differentiation, and integration. It forms the mathematical foundation of physics, engineering,

economics, and countless areas of scientific computing. In recent years, calculus has also become indispensable in artificial intelligence, where derivatives drive optimization algorithms such as gradient descent used to train deep neural networks. Although linear algebra is the primary mathematical language of quantum computing, calculus also plays an important role in areas such as quantum dynamics, differential equations, variational methods, and quantum control. SageMath provides a comprehensive environment for symbolic and numerical calculus by integrating powerful computer algebra systems and numerical libraries, making it an excellent platform for exploring both classical and modern applications of calculus.

```
sage: limit(sin(x)/x,x=0)
1
sage: diff(sin(x)^2,x)
2*cos(x)*sin(x)
sage: integral(x^2,x)
1/3*x^3
```

Figure 5: A few SageMath one-liners for calculus

Let us now understand the output produced by the calculus functions shown in Figure 5. The function `limit()` computes the limit of a function as the variable approaches a specified value. In the above example, the output 1 indicates that the limit of $\sin(x)/x$ as x approaches 0 is 1 ; this is one of the most fundamental results in differential calculus, often referred to as the first remarkable limit or the fundamental trigonometric limit. The function `diff()` performs symbolic differentiation. The output $2 \cdot \cos(x) \cdot \sin(x)$ is the derivative of $\sin^2(x)$, obtained by applying the chain rule. Finally, the function `integral()` performs symbolic integration. The output $1/3 \cdot x^3$ is the indefinite integral of x^2 , indicating that the integral of x^2 is $x^3/3$ (up to an arbitrary constant of

integration). These examples demonstrate SageMath's ability to perform symbolic calculus efficiently by leveraging powerful computer algebra systems such as Maxima, SymPy, GiNaC, Giac, and FriCAS.

Beyond the mathematical capabilities explored throughout this series, SageMath also incorporates numerous infrastructure and visualisation components, including SQLite, ZODB, Twisted, MathJax, Matplotlib, Tachyon, GD, and Jmol. These libraries provide essential services such as persistent data storage, networking, high-quality mathematical typesetting, two-dimensional and three-dimensional visualisation, rendering, and interactive user interfaces. Although they operate largely behind the scenes, they are fundamental to SageMath's architecture, transforming it from a collection of mathematical algorithms into a comprehensive ecosystem for scientific computing.

As we conclude this series, it is worth reflecting once again on why tools like SageMath matter. We live in an era where AI assistants such as ChatGPT and Claude are steadily automating many aspects of conventional programming. Writing boilerplate code and even implementing standard algorithms are becoming increasingly routine tasks. In such a world, the true value lies not merely in the ability to write code, but in understanding the mathematical and scientific principles that underpin modern computing. Scientific computing, symbolic mathematics, optimization, graph theory, cryptography, and related disciplines are therefore becoming more relevant than ever. Tools like SageMath empower us to explore these deeper ideas and transform them into working solutions. While this series may end here, the journey of scientific computing itself is only beginning, and I hope that SageMath will continue to serve as a trusted companion in your explorations. **END** 🐞

By: Dr Deepu Benson

The author is a free software enthusiast with a keen interest in theoretical computer science. He received his PhD from IIT Palakkad and completed his postdoctoral research at IIT Gandhinagar. He maintains a technical blog.